

IMAN IRAEI

M.Sc. in Electrical and Computer Engineering

iman.iraiei@mail.concordia.ca | imaniraei.github.io | LinkedIn | Google Scholar | +1-514-431-6339

RESEARCH INTERESTS

Research Interests: Computer Vision, Object Detection, Vision–Language Models, Multimodal Learning.

EDUCATION

M.Sc. – Concordia University [GPA: 4.02/4.30] <i>Gina Cody School of Engineering and Computer Science. Advised by Prof. Omair Ahmad</i>	Montreal, Canada Jan. 2025 – Present
M.Sc. – Tehran Polytechnic [GPA: 3.58/4.00] <i>Department of Electrical Engineering. Advised by Prof. Karim Faez</i>	Tehran, Iran Sep. 2013 – Feb. 2015
B.Sc. – Tehran Polytechnic [GPA: 3.44/4.00] <i>Department of Electrical Engineering. Advised by Prof. Karim Faez</i>	Tehran, Iran Sep. 2005 – Sep. 2009

RECOGNITION AND AWARDS

- Awarded Concordia University Graduate Fellowship. Jan. 2025
- Ranked top **1%** among Electrical Engineering (Electronic) co-students at Tehran Polytechnic.
- Ranked **2nd** among 32 participants in the First National Robotic Competition at IT Academy in the field of Rescue Robot. Feb. 2009
- Ranked **3rd** among 45 participants in the First National Robotic Competition at Shahrood University of Technology in the field of Sweeper Robot. Mar. 2008
- Ranked **2nd** among 104 participants in the Sixth International Robotic Competition of Intelligent Mice at Islamic Azad University of Sabzevar in the field of Line Follower Robot. Jul. 2007

RESEARCH PROJECTS

- Submitted Paper** | *LAHN-DriveCLIP: Localization-Aware CLIP Adaptation with CAM-Guided Hard Negative Mining for Autonomous Driving* May. 2026
- Developed a localization-aware vision–language framework based on CLIP and LoRA adaptation for autonomous driving applications.
 - Proposed CAM alignment and CAM-guided hard negative mining to improve weakly supervised localization.
 - Submitted to *Robotics and Autonomous Systems*.
- Submitted Paper** | *SCOPE: Spatial-Channel Optimization with Efficient Adaptive Fusion for Real-Time Object Detection in Autonomous Vehicles* June. 2026
- Proposed a latency-aware adaptive fusion framework that enhances EfficientDet with spatial-channel attention refinement and adaptive channel-wise feature weighting for autonomous driving perception.
 - Designed a single-layer adaptive attention BiFPN to prioritize informative multi-scale features while reducing redundant feature propagation and computational overhead.
 - Submitted to *IEEE Transactions on Intelligent Vehicles*.
- Poster** | *DriveCLIP: Localization-Aware CLIP for Autonomous Driving* April. 2026
- Presented at the 10th Annual ECE Graduate Student Research Conference (GSRC), Concordia University.
- Project** | *Music Genre Classification using Spectral Analysis and Random Forest Classifier* March. 2025
- Extracted MFCC-based spectral features to characterize audio signals across multiple music genres.
 - Compared standard machine learning classifiers including SVM, Random Forest, KNN, and Gradient Boosting.
 - Evaluated classification performance using accuracy measures and genre confusion matrices.
- Project** | *Classification of Chest X-ray Images using Convolutional Neural Networks* Dec. 2025
- Studied CNN-based classification of chest X-ray images for respiratory disease diagnosis.
 - Compared ResNet18, AlexNet, and MobileNetV2 on NIH, TBX11, and Kaggle Chest X-Ray datasets.
 - Investigated the effect of learning rate and batch size on classification performance.
- Journal Paper** | *A Motion Parameters Estimating Method Based on Deep Learning for Visual Blurred Object Tracking* March. 2021
- Investigated and implemented object tracking algorithms with emphasis on occlusion handling under background clutter, illumination variation, and scale changes in blurred and non-blurred scenes.

Journal Paper | *A New Approach to Enhance the Estimation of Motion Blur Parameters in Blurred Images* July. 2020

- Worked on image restoration and image retrieval projects for image deblurring.

Conference Paper | *Object Tracking in Blurred Images with Mean Shift and Wavelet Features* March. 2018

- International Conference on Pattern Recognition and Computer Vision, Rome, Italy.

Conference Paper | *Blur Parameters Estimation using SUMFC and Wavelet Transform* Nov. 2015

- International Conference on Applied Research in Computer and Information Technology, Tehran, Iran.

Conference Paper | *Object Tracking with Occlusion Handling using Mean Shift, Kalman Filter, and Edge Histogram* March. 2015

- IEEE International Conference on Pattern Recognition and Image Analysis, Guilan, Iran.

PRESENTATIONS

Object Tracking in Motion Blurred Images with Adaptive Mean Shift and Wavelet Features.

International Conference on Pattern Recognition and Computer Vision, Mar. 2018, Rome, Italy

- Main Audience: International researchers and graduate students.

Object Tracking with Occlusion Handling using Mean Shift, Kalman Filter, and Edge Histogram.

IEEE International Conference on Pattern Recognition and Image Analysis, Mar. 2015, Rasht, Iran

- Main Audience: International researchers and graduate students.

TEACHING EXPERIENCE AND DEVELOPMENT

Teaching Assistant | *ELEC 342, Discrete-Time Signals and Systems (DSP)* | *Concordia University* Fall 2025

Teaching Assistant & Marker | *ELEC 311, Electronics I* | *Concordia University* Winter 2025

Teaching Assistant | *Circuit Analysis II* | *Tehran Polytechnic* Winter 2015

Tutor & lab leader | *MATLAB Programming Principles* | *Tehran Polytechnic* Fall 2014

Teaching Assistant | *Signals and Systems* | *Tehran Polytechnic* Winter 2014

Lecturer | *Image Processing using MATLAB* | *Shariati Technical and Vocational University* Spring 2011

Lecturer | *Signals and Systems* | *Shariati Technical and Vocational University* Fall 2010

Lecturer | *Signal Processing using MATLAB* | *Tehran Institute of Technology* Fall 2009 – 2014

Lecturer | *Image Processing using MATLAB* | *Tehran Institute of Technology* Fall 2009 – 2014

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C/C++, JavaScript

Libraries and Frameworks: PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn

Developer Tools: VS Code, Jupyter Notebook, Git, Google Cloud Platform, HPC Clusters (Slurm)

REFERENCES

Omair Ahmad | *Professor, Department of Electrical and Computer Engineering at Concordia University, omair@ece.concordia.ca*

M. N. S. Swamy | *Professor, Department of Electrical and Computer Engineering at Concordia University, swamy@ece.concordia.ca*

Karim Faez | *Professor, Department of Electrical Engineering at Tehran Polytechnic, kfaez@aut.ac.ir*